中值定理

$$y = f(x)(x \in D), x_0 \in D$$

1. 若$\exists\delta > 0$, 当$0 < |x - x_0| < \delta$时,
   $f(x) < f(x_0)$
   $x = x_0$为$f(x)$的极大点
2. 若$\exists\delta > 0$, 当$0 < |x - x_0| < \delta$时,
   $f(x) > f(x_0)$
   $x = x_0$为$f(x)$的极小点

$$f'(a)\begin{cases} > 0 \\ < 0 \\ = 0 \\ \text{不存在} \end{cases}$$

1. $f'(a) > 0 : f'(a) = \lim\limits_{x \to a} \dfrac{f(x) - f(a)}{x - a} > 0$

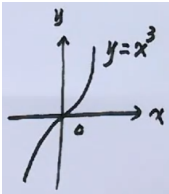
   $\exists\delta > 0$, 当$0 < |x - a| < \delta$时, $\dfrac{f(x) - f(a)}{x - a} > 0$

   $\begin{cases} f(x) < f(a), x \in (a - \delta, a) - \text{左小} \\ f(x) > f(a), x \in (a, a + \delta) - \text{右大} \end{cases} \Rightarrow x = a\text{不是极值点}$
2. $f'(a) < 0 : f'(a) = \lim\limits_{x \to a} \dfrac{f(x) - f(a)}{x - a} < 0$

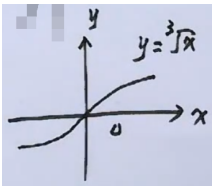
   $\exists\delta > 0$, 当$0 < |x - a| < \delta$时, $\dfrac{f(x) - f(a)}{x - a} < 0$

   $\begin{cases} f(x) > f(a), x \in (a - \delta, a) - \text{左大} \\ f(x) < f(a), x \in (a, a + \delta) - \text{右小} \end{cases} \Rightarrow x = a\text{不是极值点}$

① $x = a$为$f(x)$的极值点 $\Rightarrow (\not\Leftarrow) f'(a) = 0$或$f'(a)$不存在
② $f(x)$可导且$x = a$为$f(x)$的极值点 $\Rightarrow (\not\Leftarrow) f'(a) = 0$

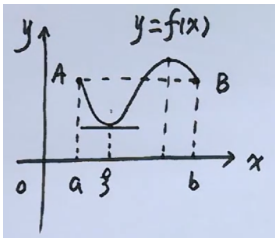


$$f(x) = x^3, f'(x) = 3x^2, f'(0) = 0$$
$$x = 0\text{不是极值点}$$



$$f(x) = \sqrt[3]{x}$$
$$\lim_{x \to 0} \frac{f(x) - f(0)}{x} = \lim_{x \to 0} \frac{\sqrt[3]{x}}{x} = \infty \Rightarrow f'(0)\text{不存在}$$
$$x = 0\text{不是极值点}$$

Rolle



若①$f(x)$在$[a,b]$上连续
②$f(x)$在(a,b)内可导
③$f(a)=f(b)$

则$\exists\xi\in(a,b)$, 使$f'(\xi)=0$

$$
\begin{aligned}
\text{证}:&1.\ f(x)\in C[a,b]\Rightarrow\exists m,M\\
&2.\ ①m=M:f(x)\equiv C_0,\forall\exists\in(a,b),\text{有}f'(\xi)=0\\
&\quad ②m<M:\\
&\qquad\because f(a)=f(b),\therefore m,M\text{至少有一个在}(a,b)\text{内取到}\\
&\qquad\text{设}\exists\xi\in(a,b),\text{使}f(\xi)=M\\
&\qquad\because\xi\text{为极大点},\therefore f'(\xi)=0\text{或}f'(\xi)\text{不存在},\\
&\qquad\because f(x)\text{在}(a,b)\text{内可导},\therefore f'(\xi)=0
\end{aligned}
$$

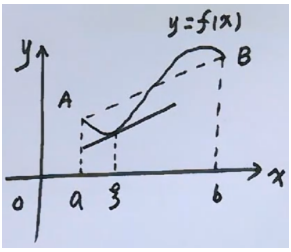
验证函数$f(x)=x^2-2x+4$在$[0,2]$上满足罗尔定理的条件，并求驻点ξ.

$$f(x)\in C[0,2],f(x)\text{在}(0,2)\text{内可导}$$
$$f(0)=f(2)=4$$
$$\exists\xi\in(0,2),\text{使}f'(\xi)=0$$
$$f'(x)=2x-2,\text{由}2\xi-2=0\Rightarrow\xi=1$$

设$f(x)$在$[0,2]$上连续, 在$(0,2)$内可导, 且$f(0)=1,f(1)+f(2)=2$, 证名：存在$\xi\in(0,2)$, 使得$f'(\xi)=0$.

$$
\begin{aligned}
\text{证}:&1.\ f(x)\in C[1,2]\Rightarrow\exists m,M\\
&\quad 2m\le f(1)+f(2)\le 2M\\
&\quad\because f(1)+f(2)=2,\therefore m\le 1\le M\\
&\quad\therefore\exists c\in[1,2],\text{使}f(c)=1\\
&2.\ \because f(c)=f(c)=1,\therefore\exists\xi\in(0,c)\subset(0,2),\text{使}f'(\xi)=0
\end{aligned}
$$

Lagrange



若①$f(x)$在$[a,b]$上连续
②$f(x)$在(a,b)内可导

$$\text{则}\exists\xi\in(a,b),\text{使}f'(\xi)=\frac{f(b)-f(a)}{b-a}$$

$$\text{分析}:L:y=f(x)$$

$$L_{AB}:y-f(a)=\frac{f(b)-f(a)}{b-a}(x-a),\text{即}$$

$$L_{AB}:y=f(a)+\frac{f(b)-f(a)}{b-a}(x-a)$$

$$\text{证}:\text{令}\Phi(x)=\text{曲}-\text{直}=f(x)-f(a)-\frac{f(b)-f(a)}{b-a}(x-a)$$

$$\Phi(x)\in C[a,b],(a,b)\text{内可导}$$

$$\because \Phi(a)=\Phi(b)=0,\therefore \exists\xi\in(a,b),\text{使}\Phi'(\xi)=0$$

$$\text{而}\Phi'(x)=f'(x)-\frac{f(b)-f(a)}{b-a}$$

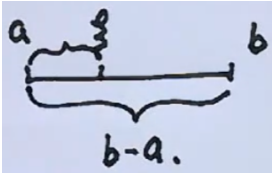
$$\therefore f'(\xi)=\frac{f(b)-f(a)}{b-a}$$

---

1. 若$f(a)=f(b),L\Rightarrow R$
2. 等价形式,

$$f(b)-f(a)=f'(\xi)(b-a)$$

$$f(b)-f(a)=f'[a+\theta(b-a)](b-a)(0<\theta<1)$$



$$\text{设}f(x)\text{二阶可导,且}f''(x)>0.\text{判断}f'(0),f'(1),f(1)-f(0)\text{的大小.}$$

1. $f(1)-f(0)=f'(c)(1-0)=f'(c)(0<c<1)$
2. $f''(x)>0\Rightarrow f'(x)\uparrow$

$$\because 0<c<1,\therefore f'(0)<f'(c)<f'(1)$$

$$\text{设函数}f(x)\text{可导,且}\lim_{x\to\infty}f'(x)=e,\text{求}\lim_{x\to\infty}[f(x+2)-f(x-1)].$$

$$f(x+2)-f(x-1)=3f'(\xi)(x-1<\xi<x+1)$$

$$\text{原式}=3\lim_{x\to\infty}f'(\xi)=3e$$

$$\lim_{x\to\infty}x^2(\sin\frac{1}{x-1}-\sin\frac{1}{x+1})(\infty*0:\frac{0}{0},\frac{\infty}{\infty})$$

$$\text{令}f(t)=\sin t,f'(t)=\cos t$$

$$\sin\frac{1}{x-1}-\sin\frac{1}{x+1}=f(\frac{1}{x-1})-f(\frac{1}{x+1})=f'(\xi)(\frac{1}{x-1}-\frac{1}{x+1})$$

$$=\frac{2}{x^2-1}\cos\xi(\frac{1}{x+1}<\xi<\frac{1}{x-1})$$

$$\text{原式}=\lim_{x\to\infty}\frac{2x^2}{x^2-1}\cos\xi=2\cos 0=2$$

$$f(x)\text{二阶可导},\lim_{x\to 0}\frac{f(x)-1}{x}=1,f(1)=2,\text{证}:\exists\xi\in(0,1),\text{使}f''(\xi)=0$$

$$\text{证}:\lim_{x\to 0}\frac{f(x)-1}{x}=1\Rightarrow f(0)=1,f'(0)=1$$

$$\exists c\in(0,1),\text{使}f'(c)=\frac{f(1)-f(0)}{1-0}=1$$

$$\because f(x)\text{二阶可导,且}f'(0)=f'(c)=1$$

$$\therefore \exists\xi\in(0,c)\subset(0,1),\text{使}f''(\xi)=0$$

Cauchy

$$\text{若①}f(x),g(x)\text{在}[a,b]\text{上连续}$$

$$\text{②}f(x),g(x)\text{在}(a,b)\text{内可导}$$

$$\text{③}g'(x)\neq 0(a<x<b)$$

$$\text{则}\exists\xi\in(a,b),\text{使}\frac{f(b)-f(a)}{g(b)-g(a)}=\frac{f'(\xi)}{g'(\xi)}$$

$$1.\ g'(x)\neq 0(a<x<b)\Rightarrow\begin{cases}g'(\xi)\neq 0\\ g(b)-g(a)\neq 0\end{cases}$$

$$2.\ \text{若}g(x)=x,C\Rightarrow L$$

$$\text{分析}:L=\Phi(x)=\text{曲}-\text{直}=f(x)-f(a)-\frac{f(b)-f(a)}{b-a}(x-a)$$

$$C:\Phi(x)=f(x)-f(a)-\frac{f(b)-f(a)}{g(b)-g(a)}[g(x)-g(a)]$$

$$\text{证}:\text{令}\Phi(x)=f(x)-f(a)-\frac{f(b)-f(a)}{g(b)-g(a)}[g(x)-g(a)]$$

$$\Phi(x)\in C[a,b],(a,b)\text{内可导},\text{且}\Phi(a)=\Phi(b)=0$$

$$\exists\xi\in(a,b),\text{使}\Phi'(\xi)=0$$

$$\text{而}\Phi'(x)=f'(x)-\frac{f(b)-f(a)}{g(b)-g(a)}g'(x)$$

$$\therefore\frac{f'(\xi)}{g'(\xi)}=\frac{f(b)-f(a)}{g(b)-g(a)}$$

$$\text{设}f(x)\in C[0,1],(0,1)\text{内可导},f(1)=0,\text{证}:\exists\xi\in(0,1),\text{使}\xi f'(\xi)+2f(\xi)=0.$$

$$\text{分析}:xf'(x)+2f(x)=0\Rightarrow\frac{f'(x)}{f(x)}+\frac{2}{x}=0$$

$$\Rightarrow[\ln f(x)]'+(\ln x^2)'=0\Rightarrow[\ln x^2f(x)]'=0$$

$$\text{证}:\text{令}\Phi(x)=x^2f(x)$$

$$\because\Phi(0)=\Phi(1)=0$$

$$\therefore\exists\xi\in(0,1),\text{使}\Phi'(x)=0$$

$$\text{而}\Phi'(x)=2xf(x)+x^2f'(x)$$

$$\therefore 2\xi f(\xi)+\xi^2f'(\xi)=0$$

$$\because\xi\neq 0,\therefore 2f(\xi)+\xi f'(\xi)=0$$

$$f(x)\in C[a,b],(a,b)\text{内可导},f(a)=f(b)=0,\text{证}:\exists\xi\in(a,b),\text{使}f'(\xi)-f(\xi)=0.$$

$$\text{分析}:f'(x)-f(x)=0\Rightarrow\frac{f'(x)}{f(x)}-1=0$$

$$\Rightarrow[\ln f(x)]'+(\ln e^{-x})'=0$$

$$\Rightarrow[\ln e^{-x}f(x)]'=0$$

$$\text{证}:\text{令}\Phi(x)=e^{-x}f(x)$$

$$\because f(a)=f(b)=0,\therefore\Phi(a)=\Phi(b)=0$$

$$\therefore\exists\xi\in(a,b),\text{使}\Phi'(\xi)=0$$

$$\text{而}\Phi'(x)=e^{-x}[f'(x)-f(x)]\text{且}e^{-x}\neq 0$$

$$\therefore f'(\xi)-f(\xi)=0$$

---

①$f(b)-f(a)$或$f(a)\neq f(b)-L$

②$\xi f'(\xi)+2f(\xi)\begin{cases}\text{仅有}\xi,\text{无}a,b\\ 2\text{项}\\ \text{导数差一阶}\end{cases}\Rightarrow xf'(x)+2f(x)\Rightarrow\frac{f'(x)}{f(x)}+\frac{2}{x}=0\Rightarrow[\ln f(x)]'+(\ln x^2)'=0\Rightarrow\Phi(x)=x^2f(x)$

③有ξ,有a,b,ξ与a,b可分开

ξ与a,b分开$\Rightarrow a,b$侧$\begin{cases}\frac{f(b)-f(a)}{b-a}-L\\ \frac{f(b)-f(a)}{g(b)-g(a)}-C\end{cases}$

$$\text{设函数}f(x)\in C[a,b],\text{在}(a,b)\text{内可导}(a>0),\text{证明：存在}\xi\in(a,b),\text{使得}f(b)-f(a)=\xi f'(\xi)\ln\frac{b}{a}.$$

$$\frac{f(b)-f(a)}{\ln(b)-\ln(a)}=\xi f'(\xi)$$

$$\text{证}:g(x)=\ln x,g'(x)=\frac{1}{x}\neq 0(a<x<b)$$

$$\exists\xi\in(a,b),\text{使}\frac{f(b)-f(a)}{\ln(b)-\ln(a)}=\frac{f'(\xi)}{\frac{1}{\xi}}\Rightarrow\ldots$$

Taylor中值定理

$$\lim_{x\to 0}\frac{x-\sin x}{x^3}\neq\lim_{x\to 0}\frac{x-x}{x^3}$$

$$\text{若}\sin x=x-\frac{x^3}{6}+o(x^3)$$

$$\text{原式}=\lim_{x\to 0}\frac{\frac{x^3}{6}-o(x^3)}{x^3}=\frac{1}{6}$$

$$\text{条件}:f(x)\text{在}x=x_0\text{邻域内有}n+1\text{阶导数}$$

$$\text{结论}:f(x)=P_n(x)+R_n(x)-\text{余项}$$

$$P_n(x)=f(x_0)+f'(x_0)(x-x_0)+\frac{f''(x_0)}{2!}(x-x_0)^2+\ldots+\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

$$R_n(x)=\begin{cases}\frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1},\xi\text{在}x_0\text{与}x\text{内}\\ o((x-x_0)^n)-\text{皮亚诺型余项}\end{cases}$$

$$\text{若}x_0=0,$$

$$f(x)=f(0)+f'(0)x+\ldots+\frac{f^{(n)}(0)}{n!}x^n+o(x^n)-\text{麦克劳林公式}$$

$$\text{记}:(x\to 0)$$

$$①e^x=1+x+\ldots+\frac{x^n}{n!}+o(x^n)$$

$$②\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\ldots+\frac{(-1)^n}{(2n+1)!}x^{2n+1}+o(x^{2n+1})$$

$$③\cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\ldots+\frac{(-1)^n}{(2n)!}x^{2n}+o(x^{2n})$$

$$④\frac{1}{1-x}=1+x+\ldots+x^n+o(x^n)$$

$$⑤\frac{1}{1+x}=1-x+x^2-\ldots+(-1)^nx^n+o(x^n)$$

$$⑥\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\ldots+\frac{(-1)^{n-1}}{n}x^n+o(x^n)$$

$$⑦(1+x)^a=1+ax+\frac{a(a-1)}{2!}x^2+\frac{a(a-1)(a-2)}{3!}x^3+\ldots$$

$$\text{求极限}\lim_{x\to 0}\frac{\ln(1+x)-e^x+1}{x^2}.$$

$$\ln(1+x)=x-\frac{x^2}{2}+o(x^2)$$

$$e^x=1+x+\frac{x^2}{2}+o(x^2)$$

$$\ln(1+x)-e^x+1=-x^2+o(x^2)\sim -x^2$$

$$\text{原式}=-1$$

$$求\lim_{x\to 0}\frac{e^{-\frac{x^2}{2}}-1+\frac{x^2}{2}}{x^2-\sin^2 x}.$$

$$e^x=1+x+\frac{x^2}{2}+o(x^2)\Rightarrow e^{-\frac{x^2}{2}}=1-\frac{x^2}{2}+\frac{x^4}{8}+o(x^4)$$

$$e^{-\frac{x^2}{2}}-1+\frac{x^2}{2}\sim\frac{x^4}{8}$$

$$原式=\frac{1}{8}\lim_{x\to 0}\frac{x}{x+\sin x}*\frac{x^3}{x-\sin x}=\frac{1}{16}\lim_{x\to 0}\frac{x^3}{x-\sin x}$$

$$\because \sin x=x-\frac{x^3}{6}+o(x^3),\therefore x-\sin x\sim\frac{1}{6}x^3$$

$$\therefore 原式=\frac{3}{8}$$

$$\lim_{x\to 0}\frac{\sqrt{1+x}+\sqrt{1-x}-2}{x^2}$$

$$由(1+x)^a=1+ax+\frac{a(a-1)}{2!}x^2+o(x^2),得$$

$$\sqrt{1+x}=1+\frac{1}{2}x-\frac{1}{8}x^2+o(x^2)$$

$$\sqrt{1-x}=1-\frac{1}{2}x-\frac{1}{8}x^2+o(x^2)$$

$$\Rightarrow\sqrt{1+x}-\sqrt{1-x}-2\sim-\frac{1}{4}x^2$$

$$\therefore 原式=-\frac{1}{4}$$